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BSCS-2A

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Abstract

With the increasing role of social media platforms in music promotion, understanding the factors that contribute to audience engagement has become essential for artists and producers. This study developed a predictive model to estimate engagement levels on YouTube music videos by integrating song characteristics and Spotify metadata. The dataset included various track attributes such as danceability, energy, valence, tempo, and instrumentalness, along with YouTube-specific features like views, likes, and comments. A Multilayer Perceptron (MLP) neural network was employed to capture the non-linear relationships between these features and engagement metrics. The model was trained using backpropagation with the Adam optimizer to minimize prediction error, achieving satisfactory convergence and generalization. Results demonstrated that combining audio features with engagement data from multiple platforms improved prediction accuracy, providing valuable insights into factors influencing music popularity. These findings have practical implications for content creators and the music industry seeking to optimize promotion strategies on digital platforms. Moreover, the trained model can be deployed in interactive

applications to provide real-time engagement predictions, assisting artists and marketers in tailoring their promotional efforts.

1 Introduction

Music consumption has undergone significant transformations with the rise of digital platforms such as YouTube and Spotify. These platforms have reshaped how audiences discover, engage with, and share music, making online engagement a crucial factor in determining a song's success. Just as recommendation algorithms have transformed how people discover movies and TV shows on streaming services, artificial intelligence and machine learning techniques have increasingly been applied to analyze and predict patterns in digital music consumption [1; 4; 7]. Online engagement in music, measured through views, likes, comments, and shares, directly impacts an artist's reach and success. Prior research highlights that social media interactions significantly influence a track's popularity and streaming performance [2; 3; 17]. However, predicting engagement levels remains a complex challenge due to the multifaceted nature of music consumption behaviors. Factors such as audio characteristics, artist popularity, and external social trends all play roles in determining a song's online reception [4; 7]. The input for this project consisted of metadata from YouTube and Spotify, including features such as a song's tempo, key, danceability, energy, and speechiness, alongside engagement metrics from YouTube videos. Previous studies have shown that combining social media analytics with audio features can improve the accuracy of music popularity predictions [17; 7]. The dataset used in

this study was sourced from Kaggle's repository, which compiles metadata from both Spotify and YouTube, providing a comprehensive set of musical and engagement-related attributes [6]. Using supervised learning, this project trained a Multilayer Perceptron (MLP) neural network to predict engagement levels based on these attributes. The output was an engagement score, providing insights into factors influencing a song's digital success.

2 Related work

This section discusses literature and empirical works relevant to this project proposal. Articles are gathered from sources such as Google Scholar, Academia, IEEE Xplore, ScienceDirect, DOAJ, Scopus, WorldCat, Springer, and ResearchGate. These referenced studies provide further background to make it easier to read and understand the purposes and primary elements of the research being presented. Recent research has explored various machine learning approaches to predicting music popularity, leveraging different types of features such as audio characteristics, social media engagement, and sentiment analysis. Rompolas et al. (2024) demonstrate how integrating textual emotion analysis with machine learning models improves the prediction of song success, highlighting the role of audience engagement in determining popularity [9]. Similarly, Sebastian, Jung, and Mayer (2024) examine multiple machine learning models, including Ordinary Least Squares (OLS), Multivariate Adaptive Regression Splines (MARS), Random Forest, and XGBoost, to analyze song features and their impact on success, finding that Random Forest

achieves the highest accuracy [10]. However, while these models are effective at capturing feature relationships, they often rely on hand-crafted features and do not incorporate dynamic user interactions in real-time. Extending this approach, Yee and Raheem (2022) introduce social media engagement from YouTube videos as an additional predictive factor, demonstrating that incorporating these features improves model performance by 10% to 60% compared to using audio data alone [11]. A limitation of this study is its focus on aggregated engagement metrics rather than temporal changes in engagement over time. Building on these findings, Dimolitsas, Kantarelis, and Fouka (2023) analyze nearly 18,500 songs and extract audio features using the Spotify Web API. This study focuses on hit song prediction using Random Forest and Support Vector Machines (SVMs), achieving an accuracy of approximately 86% [15]. While this demonstrates the effectiveness of traditional machine learning models, the study does not incorporate external engagement metrics beyond Spotify. In contrast, Khan et al. (2022) investigate the impact of feature selection on music popularity classification, concluding that reducing features can improve computational efficiency with minimal accuracy loss [12]. While feature selection enhances interpretability, it may overlook complex feature interactions that could contribute to predictive performance. Beyond traditional machine learning, Bevec, Tkalciˇ c, and Pesek (2024) explore graph-based ap- ˇ proaches, demonstrating that Graph Neural Networks (GNNs) outperform traditional collaborative filtering methods in music recommendation tasks [8]. These models excel at capturing relational data between users and tracks, but their reliance on structured graph representations makes them less adaptable to new or emerging trends driven by viral content. Some studies have

started exploring multimodal approaches, such as context-dependent sentiment analysis in user-generated videos [14] and multimodal emotion classification for music videos [13], both of which show promise in improving affective computing. However, these approaches primarily focus on content-based analysis and do not directly integrate user engagement metrics, making them less applicable for predicting online popularity. Deep learning techniques have also been explored for music popularity prediction. Rusconi (2024) applies a neural network model trained on Spotify data, demonstrating that deep learning can capture non-linear relationships between song features and popularity [16]. However, the study acknowledges challenges in hyperparameter tuning and overfitting due to limited training data. Meanwhile, Sardana (2024) proposes an ensemble learning approach combining multiple machine learning models to improve predictive accuracy, suggesting that hybrid techniques may be a promising direction for future research [17]. Similarly, Velmuruga (2023) provides a case study in music analytics, evaluating different machine learning approaches for song popularity prediction [18]. The study highlights that incorporating historical trends and streaming platform metadata significantly improves predictive performance. Compared to previous works, this project integrates both audio-based features and engagement data from YouTube and Spotify, utilizing a Multilayer Perceptron (MLP) neural network to predict music popularity. Unlike studies that focus solely on song characteristics or recommendation systems, this approach considers both the musical content and user interaction patterns. While prior studies have demonstrated that social media data can enhance prediction accuracy, they often overlook how user engagement evolves over time—a critical factor in a song's viral success. By

incorporating data from both platforms, this project aims to develop a more accurate and current model for predicting online music engagement.

3 Methods

3.1 Model Implementation

This study employed machine learning techniques to predict engagement levels of YouTube music videos based on metadata features. The primary model used was a Multilayer Perceptron (MLP) neural network, which was trained using supervised learning. The model was designed to analyze numerical features extracted from the dataset and identify patterns associated with audience engagement.

3.2 Multilayer Perceptron (MLP) Model The Multilayer Perceptron (MLP) is a feedforward neural network consisting of multiple layers, including hidden layers and an output layer. The hidden layers utilized activation functions such as ReLU, which is widely used in deep learning tasks due to its efficiency in handling non-linearity [22]. The MLP model was implemented using TensorFlow's Keras API, which provided a high-level interface for building and training neural networks, and was compatible with the available hardware.

3.3 Optimization and Training The model was trained using backpropagation with gradient descent. The optimization algorithm chosen was Adam, and the loss function

used was Mean Squared Error (MSE), reflecting the regression formulation of the engagement prediction task [20]. Hyperparameter tuning, such as adjustments to the number of layers, neurons per layer, and learning rate, was conducted based on validation results.

3.4 The Dataset The dataset used in this study was the Spotify and YouTube Dataset [24], which compiled metadata from both platforms. It included statistics on the top 10 songs of various Spotify artists and their corresponding YouTube music videos. The dataset contained 26 variables, which were categorized into different groups. The Track Information category included Track name (Track), Artist (Artist), Album (Album), Album type (Album_type), and links to Spotify (Url_spotify, Uri) and YouTube (Url_youtube). The Audio Attributes category consisted of Danceability (Danceability), Energy (Energy), Key (Key), Loudness (Loudness), Speechiness (Speechiness), Acousticness (Acousticness), Instrumentalness (Instrumentalness), Liveness (Liveness), Valence (Valence), and Tempo (Tempo). The Engagement Metrics category included Spotify streams (Stream), YouTube views (Views), Likes (Likes), Comments (Comments), and whether the video was the official version (official_video). Lastly, the Additional Features category covered Track duration (Duration_ms), Channel name (Channel), Video description (Description), and Licensing information (Licensed).

Figure 1: Sample data from the dataset

3.5 Data Preprocessing Data preprocessing involved handling missing values by imputing means for numerical features and modes for categorical features, following approaches used in prior music popularity studies [11]. Irrelevant or redundant columns, such as URLs, channel information, and track identifiers, were removed to streamline the dataset and focus on predictive variables, consistent with feature selection recommendations to improve computational efficiency and model interpretability [12]. Numerical features, including Loudness, were normalized using Min-Max scaling to standardize their ranges, similar to normalization techniques in deep learning models for music analytics [16; 18]. Categorical variables such as Album_type were encoded using factorization to convert them into numerical format suitable for neural network modeling. Highly correlated features with correlation coefficients above 0.8 were identified and dropped to reduce multicollinearity, aligning with standard preprocessing practices in machine learning [10]. In the music industry, engagement metrics like views and streams serve as critical indicators of a song's popularity and reach, with platforms such as YouTube and Spotify playing pivotal roles in artist visibility and audience growth [25; 26]. To leverage these indicators for prediction, a composite engagement score was engineered by combining views and streams with weighted coefficients. This score serves as the target output variable for the neural network model and is computed as: Score = (0.8 × V iews) + (0.2 × Stream) The weighting reflects the greater influence of views on a song's popularity and reach, consistent with industry insights [27]. Using this composite score as the output enables the model to predict overall engagement more holistically than relying on any single metric.

3.6 Model Training and Evaluation The neural network model was trained using supervised learning, leveraging numerical and categorical features extracted from the dataset. The input features included Album Type, Danceability, Energy, Key, Loudness, Speechiness, Acousticness, Instrumentalness, Liveness, Valence, Tempo, Duration (in milliseconds), and Comments. The target output was the Score, representing the song's overall engagement metric. Notably, Views and Stream counts were excluded from the inputs to avoid data leakage. To develop and evaluate the model, the dataset was randomly split into training and testing subsets following standard procedure, with 80% of the data used for training and 20% reserved for testing.The model architecture consisted of five hidden layers, each containing 1000 neurons with ReLU activation functions, followed by a single output neuron with a linear activation function to predict continuous Score values. This deep architecture allowed the model to capture complex nonlinear patterns within the data. The neural network was trained using the Adam optimizer with a learning rate of 0.001, a batch size of 32 (the default setting), and for 50 epochs. These hyperparameters were chosen based on empirical tuning and insights from prior studies on music popularity prediction and neural network training [16; 18; 21]. The selected learning rate balanced convergence speed and stability, while the batch size optimized computational efficiency without sacrificing model generalization. Epochs were set to allow sufficient training without overfitting, as indicated by validation loss monitoring. While some prior studies applied regularization techniques such as dropout and L2 regularization to mitigate overfitting and improve generalization [16; 21], this study opted not to include

these methods, considering the dataset size and preliminary training results. This approach aligns with Rusconi (2024), who highlighted challenges in hyperparameter tuning in similar neural network configurations [16]. Model performance was evaluated using regression metrics suitable for continuous prediction tasks. Mean Squared Error (MSE), Mean Absolute Error (MAE), R-squared (R2), and Explained Variance Score were calculated to assess prediction accuracy and variance explained, consistent with related research [18; 15]. Classification metrics such as accuracy, precision, and recall were considered for alternative problem formulations, as recommended in music popularity prediction studies [11; 18].

3.7 Model Evaluation Model training progress was monitored by plotting training and validation loss and mean absolute error (MAE) across epochs, providing insight into convergence behavior and potential overfitting [16]. After training, the model was saved as a .h5 file named model.h5 to preserve the trained weights and architecture for future use. The model's performance was then evaluated on the test set using regression metrics such as Mean Squared Error (MSE), Mean Absolute Error (MAE), R-squared (R2), and Explained Variance Score to assess prediction accuracy and variance explained [18; 15]. Additionally, descriptive statistics including minimum, maximum, mean, and median scores were computed to analyze general song performance trends. These insights serve as a reference for interpreting the model's predictions in real-world scenarios, such as evaluating how well a track might perform on a deployed application.

3.8 Interactive Prediction Interface To enable practical deployment of the trained Multilayer Perceptron (MLP) model, an interactive command-line interface (CLI) was implemented. This interface allows users to input a song title and retrieve both the actual and predicted engagement scores generated by the model. During preprocessing, the dataset was curated to retain song titles for reference, while non-essential features were excluded. Upon receiving user input, the program searches for the corresponding song features and returns the predicted engagement score, offering a user-friendly means to evaluate the model's output on a per-song basis. This implementation illustrates the model's applicability in real-time scenarios, extending its utility beyond static, offline analysis.

4 Results and Discussion

4.1 Experimental Setup and Hyperparameter Selection The model's training progress, shown by steadily decreasing loss and MAE on both training and validation sets (see training logs), indicates effective convergence. Validation metrics remain stable without divergence from training metrics, suggesting that the selected hyperparameters supported learning without signs of overfitting.

4.2 Evaluation Metrics Performance was evaluated using the following metrics: • Mean Squared Error (MSE) to measure regression accuracy. • Mean Absolute Error (MAE) to assess average prediction error. • R-squared (R2) to quantify variance explained.

4.3 Quantitative Results Table 1 summarizes the model's performance on the test set. Figure 2 displays training and validation loss and MAE over epochs, indicating convergence behavior. Descriptive statistics of the target engagement Score are summarized in Table 2. These values reflect the distribution range and skewness of the dataset, providing insight into the variability of the target engagement Score. Metric MSE MAE R2 Explained Variance Value 0.000570 0.011665 0.584316 0.595103 Table 1: Performance Metrics on Test Data Statistic Value Minimum Score 4.42 ×10−8 Maximum Score 0.8890 Mean Score 0.0173 Median Score 0.0061 Table 2: Descriptive Statistics of Target Scores

Figure 2: The figure displays the training and validation loss (left) and Mean Absolute Error (MAE) (right) across 50 epochs. Both loss and MAE show a consistent downward trend, indicating that the model is learning effectively over time. The training and validation curves closely follow each other, suggesting that the model is not overfitting and is generalizing well to unseen data. Although the validation metrics exhibit some fluctuation, the overall convergence pattern is clear. This behavior indicates that the model's performance stabilizes after approximately 30–40 epochs.

4.3.1 Collinearity Analysis To assess multicollinearity among features, a correlation matrix heatmap was generated. The analysis revealed that the ‘Likes‘ feature exhibited high correlation with ‘Views‘ (correlation coefficient > 0.8) and was therefore removed during preprocessing to reduce redundancy and stabilize model training. Figure 3: Correlation matrix heatmap of input features. Strongly correlated variables (above 0.8) were excluded.

4.4 Qualitative Results To further assess the model’s predictive behavior, Figure 4 presents a scatter plot comparing actual and predicted engagement scores. The plot reveals that while the model performs adequately for lower engagement scores, its predictions become less consistent at higher score levels. This behavior may be attributed to the skewed distribution of the target variable or limitations in capturing complex engagement patterns at the upper extremes. Examples of model predictions alongside actual values are shown in Figure 3. These examples illustrate that the model tends to perform better on songs with high engagement scores, while less popular songs exhibit greater prediction variance, likely reflecting the noisier and less consistent patterns present in the lower range of the dataset. Song ID Actual Score Predicted Score 150 0.001293 0.008347 297 0.000524 0.008430 1583 0.002838 0.008287 1611 0.001014 0.006011 1645 0.161280 0.090479 1967 0.003109 0.007878 2352 0.003118 0.008741 2568 0.002799 0.008153 Table 3: Examples of model predictions alongside actual values for selected songs.

Figure 4: Scatter plot comparing actual and predicted engagement scores for songs using the trained neural network model. Each point represents a song's actual (x-axis) and predicted (y-axis) score. The red dashed line indicates the line of perfect prediction (y = x), where the model's output exactly matches the true value. Points near this line indicate accurate predictions, while deviations highlight model error. The plot suggests reasonable predictive alignment in lower score ranges, though variance increases with higher actual scores, potentially due to data imbalance or model overfitting.

4.5 Interactive Prediction Interface To demonstrate the practical applicability of the developed neural network model, a command-line interface (CLI) was implemented to enable real-time prediction of engagement scores for specific songs. Unlike the preprocessing performed during model training, the dataset used in this interface retains the Track (song name) column to facilitate lookup by song title. The interface preprocesses the dataset by filling missing values, encoding categorical features, normalizing numerical features using Min-Max scaling, and removing highly correlated input variables to reduce redundancy, consistent with the training pipeline. A composite engagement Score is calculated as a weighted sum of Views and Streams (80% and 20%, respectively), with the original components subsequently removed from the input features. Upon receiving a song name input (case-insensitive), the CLI searches the dataset for an exact match, retrieves the corresponding feature vector (excluding Track and Score), and uses the pre-trained Multilayer Perceptron (MLP) model to predict the engagement score. The actual and predicted scores are then

displayed to the user. This process repeats until the user opts to exit. The CLI provides an interactive means for users to evaluate model predictions on-demand, highlighting a pathway for integrating the predictive model into accessible applications such as music analytics dashboards or promotional tools. Code excerpt illustrating the prediction function: def predict_song_score(song_name): mask = df['Track'].str.lower() == song_name.lower() if not mask.any(): print(f"Song '{song_name}' not found.") return song_data = df[mask] actual_score = song_data['Score'].values[0] X = song_data.drop(['Score', 'Track'], axis=1) pred_score = model.predict(X) print(f"Actual Score: {actual_score:.4f}") print(f"Predicted Score: {pred_score[0][0]:.4f}") Future work may involve expanding this interface into a web-based or mobile application to enable broader accessibility and real-time engagement prediction for new releases. 4.5.1 Example Predictions from the Command-Line Interface To demonstrate the practical utility of the predictive model, a command-line interface (CLI) was implemented that allows users to input a song name and obtain both the actual and predicted engagement scores. Table 4 presents example outputs from this interface for several popular songs included in the dataset. Song Actual Engagement Score Predicted Engagement Score Take on me 0.2354 0.1552 In the end 0.2326 0.1408 Believer 0.3746 0.1819 Bad guy 0.2514 0.2176 Shape of you 0.7836 0.3510 Table 4: Example actual vs. predicted engagement scores from the CLI for selected popular songs. Notably, some well-known songs such as "Thriller" and "Bad Blood" were not found in the dataset, reflecting dataset limitations in coverage. The predicted scores generally follow the trend of actual engagement but tend to underestimate higher engagement levels, consistent with earlier quantitative observations. This CLI tool thus offers an

interactive means to explore the model's performance on specific tracks, which could be further developed into a user-friendly application for music industry stakeholders.

4.6 Discussion

The results indicate that the Multilayer Perceptron (MLP) neural network effectively predicts music engagement scores by integrating audio-based features with user engagement metrics from both YouTube and Spotify. The model achieved a moderate coefficient of determination (R2 = 0.58), demonstrating its capacity to explain a substantial portion of the variance in engagement data. This finding corroborates previous research by Velmuruga (2023) and Dimolitsas et al. (2023), who reported that the inclusion of historical trends and audio features enhances predictive performance [18; 15]. The present findings further support the advantage of combining multiple data modalities. While traditional machine learning methods such as Random Forests and Support Vector Machines (SVMs) have demonstrated efficacy when applied solely to audio features [15; 12], the use of an MLP model incorporating engagement data reflects the benefits described by Yee and Raheem (2022), who integrated social media metrics to improve prediction accuracy [11]. In contrast to prior studies that primarily utilized aggregated engagement metrics, this work employed a composite engagement score designed to capture a more comprehensive representation of audience interaction, consistent with insights from industry engagement analyses [27]. The variability observed in prediction accuracy, particularly at higher engagement levels, underscores the challenges associated with modeling viral dynamics and

evolving user interactions. This limitation is consistent with the observations of Sebastian et al. (2024), who noted that models relying on hand-crafted features may be insufficient to fully capture the complexity and dynamism of popularity patterns [10]. Similarly, Bevec et al. (2024) highlighted the difficulties faced by graph-based models in adapting to emerging trends driven by viral content [8]. Notably, the deep learning approach implemented herein demonstrated stable convergence without evidence of overfitting, despite the relatively deep network architecture and absence of explicit regularization techniques. This outcome aligns with the findings of Rusconi (2024), who emphasized the capacity of neural networks to model non-linear relationships in music data when appropriate hyperparameter tuning is applied [16]. Nonetheless, future research could investigate the incorporation of regularization methods and temporal engagement features to enhance model robustness and predictive accuracy. Furthermore, the developed prediction pipeline has practical potential beyond academic analysis. By integrating the trained Multilayer Perceptron model into a user-friendly application, such as the command-line interface developed in this study, artists, producers, and marketers could obtain real-time predictions of a song's engagement potential based on its audio features and metadata. Such an app could assist content creators in optimizing promotional strategies, tailoring releases, and monitoring evolving audience interaction trends across platforms. Future work could focus on developing this deployment, including enhancing model robustness with live data and incorporating additional social media metrics to improve prediction accuracy and applicability. In summary, this study contributes to the expanding literature on multimodal and hybrid approaches to music popularity prediction. By leveraging both

audio characteristics and dynamic engagement metrics from multiple platforms, the model provides a more comprehensive framework with potential applications in music analytics, recommendation systems, and marketing strategies.

5 Summary

This study sought to develop a predictive model for online music popularity by integrating audiobased features with user engagement metrics sourced from both Spotify and YouTube platforms. The research objective focused on enhancing the accuracy of music popularity prediction through the application of a Multilayer Perceptron (MLP) neural network, designed to capture complex nonlinear relationships within multimodal data. The results demonstrate that the MLP model effectively predicts a composite engagement score, attaining an R2 value of 0.58 and accounting for a substantial portion of the variance in user engagement. These findings corroborate prior research underscoring the importance of combining audio features with streaming and social media metrics to improve predictive performance [15; 18; 11]. The observed stability in training and validation curves indicates successful model convergence without overfitting, consistent with previous studies involving neural networks in music analytics [16]. While conventional algorithms such as Random Forests and Support Vector Machines have demonstrated strong performance in music hit prediction tasks, the deep learning approach employed in this study benefits from its ability to model nonlinearities and incorporate diverse feature types, yielding competitive results. Nevertheless, prediction accuracy declined for songs exhibiting

very high engagement scores, likely attributable to skewed data distributions and the inherent complexity of viral phenomena, reflecting challenges identified in existing literature [10; 8]. Future research directions include expanding the dataset to incorporate temporal engagement patterns and exploring the implementation of additional regularization techniques, such as dropout or L2 regularization, to enhance model robustness. Moreover, investigating ensemble or hybrid models that combine neural networks with graph-based or multimodal sentiment analysis approaches may better capture complex user interactions and emerging trends. Enhancing computational resources and utilizing larger, more diverse datasets could facilitate the development of deeper network architectures and more refined hyperparameter optimization. In conclusion, this study contributes to the field by demonstrating the efficacy of a multimodal, neural network–based approach to music popularity prediction. The findings hold practical significance for applications in music analytics, recommendation systems, and digital marketing strategies within the evolving music industry. Additionally, the model's integration into user-friendly applications offers promising avenues for real-time engagement prediction, potentially benefiting artists, producers, and marketers in optimizing their content promotion strategies.